# Exit Ticket · reflect and retrieve

## Exit Ticket

True or false: an else branch runs when its if condition is false.

## Use key vocabulary from memory

1. Explain sequence in your own words.

2. Explain selection in your own words.

3. Explain Boolean in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 038 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Programming Techniques · Sequence and Selection

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does a full adder include that a half adder does not?

### Q2

What is a local variable?

### Q3

Why does a stack use LIFO?

### Q4

Simplify A ∨ (A ∧ B).

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Explain the important decision — Sequence changes score before selection tests it. The condition is true at exactly 10 because >= includes the boundary.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Apply independently

Explain one difference between sequence and selection. [2]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Apply independently

Trace x = 4; x = x + 7; output x. [2]

## Question 2

Apply independently

Write a C# condition that prints Adult when age >= 18. [3]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

sequence — Instructions run in order

selection — A conditional branch

Boolean — True or false expression

condition — A logical test

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Follow the order of execution

## Follow the worked method

1. Update in sequence — score starts at 7. Adding 3 changes the same variable to 10; the condition has not been tested yet.

2. Evaluate the condition — Substitute the new value: 10 >= 10 is true. Using the old value 7 would trace the wrong state.

3. Take the selected branch — The true branch prints Pass. For initial scores 6, 7 and 8, the updated scores are 9, 10 and 11: only the latter two print Pass.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Discuss, then respond
value = 3; value *= 4; is value > 10?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.